GCSE GEOGRAPHY

Paper 2: Challenges in the human environment

Topic 4: Urban issues and challenges

Practice questions in the style of AQA GCSE Geography (8035). Every question is original; the layout, command words, mark allocations (including levels-based 6- and 9-mark questions with SPaG) and mark scheme follow the real exam format. Figures and data are illustrative and simplified for practice; case-study facts in the mark scheme are summarised, so check them against your own case-study notes.

Time allowed	About 110 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a black pen, a pencil, a ruler, a calculator and (for some questions) a protractor.
Instructions	• Use black ink or black ball-point pen. Use pencil only for drawing. • Answer **all** questions. • You must answer the questions in the spaces provided. • Do all rough work in this book. Cross through any work you do not want to be marked. • Questions that carry a **SPaG** note are also marked for spelling, punctuation, grammar and the use of specialist terminology.
Information	• The maximum mark for this paper is **105**. • The marks for questions are shown in brackets. • Questions that start with the command word *Explain* need developed points (a point plus a reason or consequence). • 6- and 9-mark questions are marked by **levels**: aim to give a detailed, balanced and well-linked answer, using place-specific detail.

Name: _____ Date: _____ Mark: _____ / 105

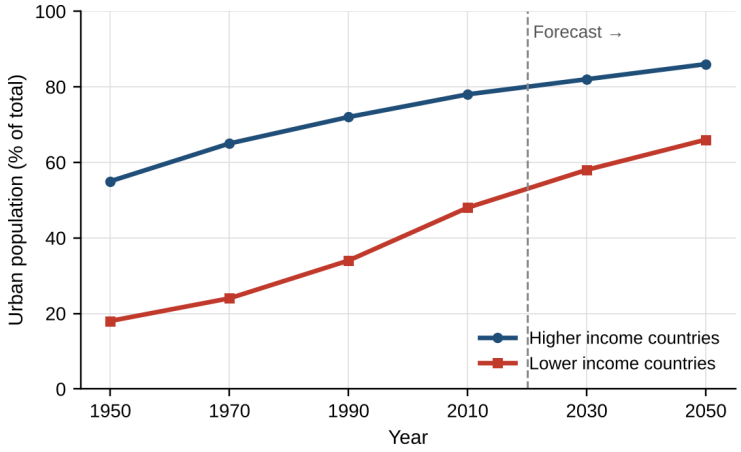
Section A: Urban issues and challenges

Answer all questions in this section.

0 1 This question is about urban growth. **Figure 1** shows the percentage of the population living in urban areas.

0 1 . 1 What was the percentage of people living in urban areas in lower income countries in 1990? **[1 mark]**

Figure 1

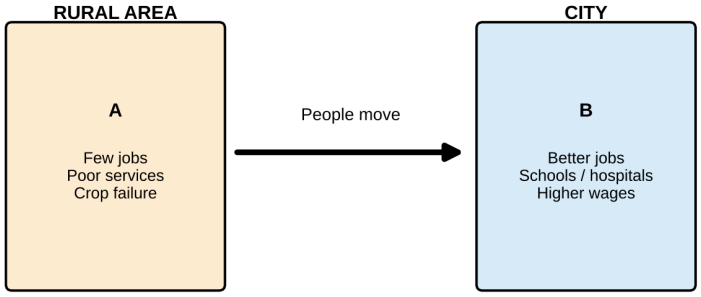


0 1 . 2 Describe **two** differences between the trends for higher income and lower income countries shown in Figure 1. **[2 marks]**

0 1 . 3 State what is meant by **urbanisation**. **[1 mark]**

0 1 . 4 **Figure 2** shows push and pull factors. Give **one** push factor and **one** pull factor that cause people to move from rural to urban areas in a lower income country. **[2 marks]**

Figure 2



0 1 . 5 Explain how **natural increase** contributes to urban growth in cities in lower income countries. **[3 marks]**

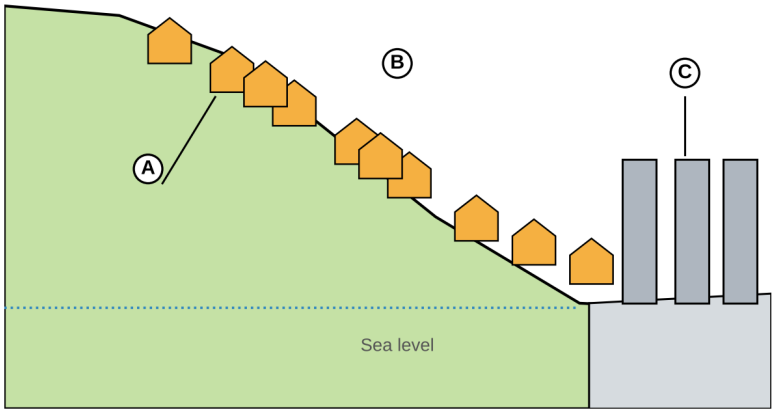
0 1 . 6 Explain why the number of megacities is increasing in lower income countries. **[3 marks]**

0 2 This question is about a major city in a lower income or newly emerging country (LIC / NEE). Refer to **Rio de Janeiro** (or another city you have studied).

0 2 . 1 Give **two** reasons why Rio de Janeiro is important at a national or international scale. **[2 marks]**

0 2 . 2 **Figure 3** shows part of a squatter settlement (favela) in Rio. Describe **two** features of the housing shown. **[2 marks]**

Figure 3



0 2 . 3 Explain **two** problems faced by people living in squatter settlements (favelas) in Rio de Janeiro. **[4 marks]**

0 2 . 4 Describe **one** social and **one** economic opportunity created by urban growth in Rio de Janeiro. **[4 marks]**

0 2 . 5 Explain how **one** environmental challenge has been caused by urban growth in Rio de Janeiro or another city you have studied. **[3 marks]**

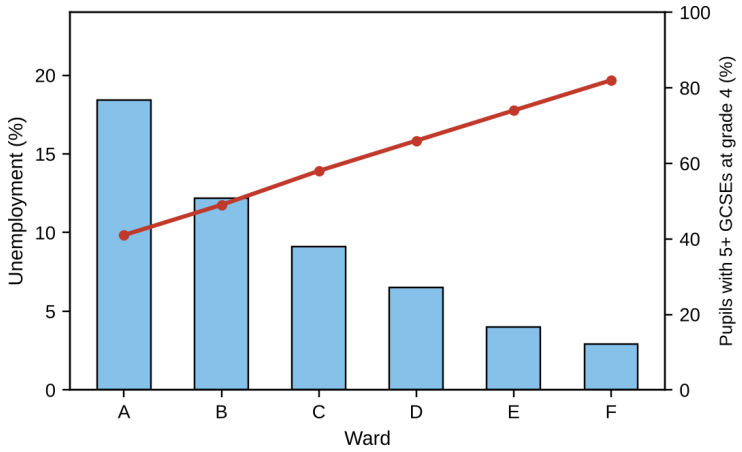
0 2 . 6 Describe the main features of an urban planning scheme that has been used to improve the quality of life of the urban poor in Rio de Janeiro (e.g. Favela Bairro). **[4 marks]**

0 2 . 7 Evaluate the success of an urban planning scheme to improve the lives of poor people in a city in an LIC or NEE. **[6 marks]**

0 3 This question is about urban change in the UK. **Figure 4** shows deprivation data for six wards in a UK city.

0 3 . 1 Describe the relationship between the percentage of unemployment and the educational performance shown in Figure 4. **[3 marks]**

Figure 4



0 3 . 2 Calculate the range in unemployment rate between the wards shown in Figure 4. **[2 marks]**

Answer = _____ %

0 3 . 3 Suggest **two** reasons why unemployment is higher in some wards than others in the same UK city. **[4 marks]**

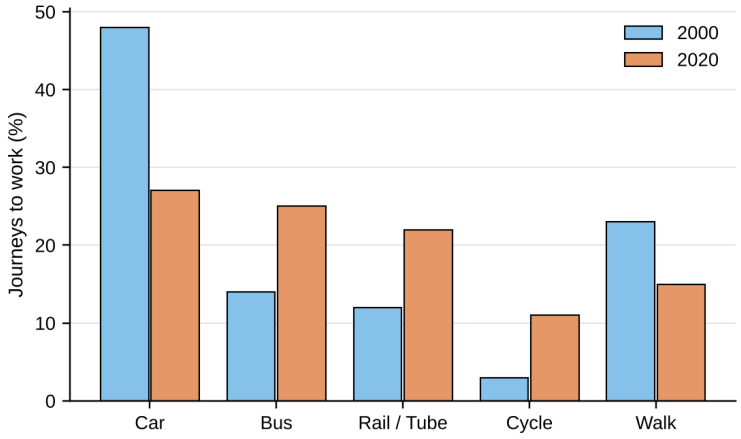
0 3 . 4 Describe **two** ways in which a UK city you have studied (e.g. London) is important in the UK and the wider world. **[2 marks]**

0 3 . 5 Explain how **migration** has changed the character of a UK city you have studied. **[4 marks]**

0 3 . 6 Describe how **deindustrialisation** and **decentralisation** have changed urban areas in the UK. **[4 marks]**

0 3 . 7 **Figure 5** shows how people travel to work in a UK city in 2000 and 2020. Describe **two** changes shown. **[2 marks]**

Figure 5

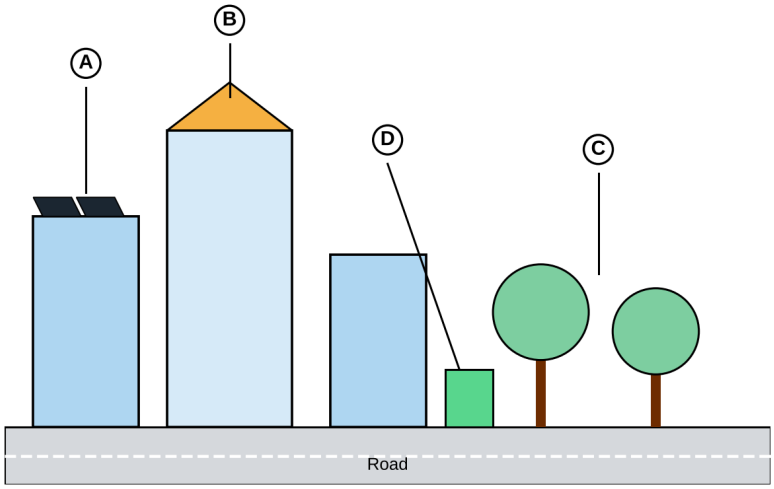


0 3 . 8 Explain how an urban regeneration project in a UK city you have studied (e.g. Stratford, London: the Olympic Park) has improved the area. **[6 marks]**

0 4 This question is about sustainable urban living and managing urban transport. **Figure 6** shows features of a sustainable urban area.

0 4 . 1 Name feature **A** and feature **C** in Figure 6. **[2 marks]**

Figure 6



0 4 . 2 State what is meant by **sustainable urban living**. **[1 mark]**

0 4 . 3 Describe **two** ways in which water and energy conservation can make a city more sustainable. **[4 marks]**

0 4 . 4 Explain how creating **green space** can improve the quality of life in a city. **[3 marks]**

0 4 . 5 Explain how **two** strategies could reduce traffic congestion in a city. **[4 marks]**

0 4 . 6 **'Reducing the use of private cars is the best way to make cities more sustainable.'** To what extent do you agree? **[9 marks] + [3 marks] SPaG**

Spelling, punctuation, grammar and use of specialist terminology [3 marks]

0 5 This question is about urban challenges in the UK. **Table 1** shows data about housing in four UK cities.

0 5 . 1 Which city has the highest average house price to earnings ratio? **[1 mark]**

Table 1

	City A	City B	City C	City D
Average house price (£)	215 000	560 000	180 000	330 000
Average annual earnings (£)	31 000	42 000	28 000	34 000
Households on housing waiting list (%)	6	14	5	9

0 5 . 2 Calculate the house price to earnings ratio for City D. **[2 marks]**

Answer = _____

0 5 . 3 Suggest **two** reasons why house prices are higher in City B than in City C. **[2 marks]**

0 5 . 4 Explain **two** ways in which new housing can be provided in a UK city without building on greenfield land. **[4 marks]**

0 5 . 5 Explain why **social deprivation** is often higher in inner city areas than in the suburbs. **[4 marks]**

0 5 . 6 Describe **one** way in which a city's CBD (central business district) has changed in recent years. **[2 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4: Urban issues and challenges

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *geography*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	34 (%)	Allow 33–35.	1	AO3 3.2.1.1
01.2	higher income countries have a much higher % urban than lower income countries throughout (e.g. 78% vs 48% in 2010) ; lower income countries are growing faster / change is greater (18% → 48%, +30, vs 55% → 78%, +23, between 1950 and 2010) ; the gap between them narrows over time (37 points in 1950 → 20 points in 2050) ;	Any two comparisons, with data for 1 mark each.	2	AO3 3.2.1.1
01.3	an increase in the proportion of people living in towns and cities / urban areas		1	AO1 3.2.1.1
01.4	push: few jobs / low wages / poor services / drought or crop failure / land shortage ; pull: better paid jobs / schools and hospitals / improved quality of life / family already there ;		2	AO2 3.2.1.1
01.5	migrants are often young adults of child-bearing age ; so they have children in the city, increasing the birth rate ; death rates in cities fall because of better healthcare, so the population grows ;		3	AO2 3.2.1.1
01.6	rapid rural-to-urban migration ; high rates of natural increase in the city population ; economic growth: globalisation / industries / TNCs locating in cities, attracting more people ;		3	AO2 3.2.1.1
02.1	regional capital / former capital of Brazil ; major port and airport / transport hub ; hosts international events (2014 World Cup, 2016 Olympic Games, Carnival) ; centre of media / TV, finance / oil industries, headquarters of large companies ; tourism – Christ the Redeemer, Copacabana, Sugar Loaf ;	Any two. Must be specific to the named city.	2	AO1 3.2.1.2
02.2	built on steep hillside slope ; close together / densely packed / crowded ; informal / self-built from cheap materials such as brick, wood and corrugated iron ; close to planned tower blocks / next to wealthy areas ;	Allow features seen in the sketch. A = informal housing; C = formal tower blocks.	2	AO3 3.2.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	no formal rights to the land, so risk of eviction ; no clean water / sewage / electricity, so health problems / disease spread ; built on steep slopes: landslides in heavy rain ; high crime rates / drug gangs / violence ; unemployment / informal low-paid jobs ;	Two problems, each with 1 mark for the problem and 1 for the development.	4	AO2 3.2.1.2
02.4	social: access to schools, hospitals and better healthcare than in rural areas / improved quality of life ; more electricity / water supply / services for many (e.g. Favela Bairro) ; economic: formal jobs in industry (e.g. oil, steel, tourism), services ; informal economy: recycling, selling goods; many small businesses ; money sent back to rural areas as remittances / growth in the economy ;		4	AO1 3.2.1.2
02.5	traffic congestion and car use leading to air pollution ; waste disposal: much rubbish in streets / rivers / Guanabara Bay polluted ; untreated sewage entering rivers and the bay, harming wildlife and spreading disease ;		3	AO2 3.2.1.2
02.6	named scheme e.g. Favela Bairro (1990s–) ; residents given legal land ownership / rights to stay ; provision of clean water, sewers, electricity, paved streets, and rubbish collection ; building of community facilities: schools, health clinics, play areas ; self-help: residents help build / improve their own homes (e.g. Rocinha), so schemes are cheap and sustainable ;		4	AO1/AO2 3.2.1.2
02.7	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO3 3.2.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
	Indicative content (credit any other valid geography, and place-specific detail): • named scheme (Favela Bairro, Rocinha) and aims ; • successes: improved sanitation, water and electricity for thousands of residents ; roads and public spaces improved ; increased land value ; • social improvements: feeling of safety / less disease / lower child death rates ; • limitations: expensive, covers only some of the favelas / only a small proportion are reached ; high maintenance cost ; • gentrification and evictions ; crime and gangs not tackled ; • reasoned judgement e.g. 'successful for those reached but not enough'. ;			
03.1	as unemployment falls from ward A to ward F, the percentage of pupils with 5+ GCSEs at grade 4 rises ; negative relationship / inverse ; ward A has the highest unemployment (18.4%) and lowest attainment (41%) ; ward F the lowest unemployment (2.9%) and highest attainment (82%) ;		3	AO3 3.2.1.3
03.2	18.4 − 2.9 ; 15.5 (%) ;		2	AO4 3.2.1.3
03.3	loss of traditional industries (e.g. manufacturing, docks) left inner areas with few jobs ; low skills / fewer qualifications ; wealthier people moved to suburbs near better jobs and schools ; poor transport links to jobs ; housing costs / quality, deprived neighbourhoods have a poor image, deterring new employers ;		4	AO2 3.2.1.3
03.4	capital city / seat of government ; global financial centre (the City of London) / HQ of TNCs ; diverse culture / many universities / tourism / theatres ; major transport hub (Heathrow, rail) ;		2	AO1 3.2.1.3
03.5	international migration brought many cultures / ethnic groups (e.g. Brick Lane, Southall) ; wide range of food, festivals and religious buildings (cultural mix) ; young population (migrants), so more births / natural increase ; pressure on housing / schools / services but also workers for NHS and the economy ; rural-to-urban migration of young workers within the UK ;		4	AO2 3.2.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	deindustrialisation: decline of manufacturing / old factories and docks closed, leaving brownfield sites and unemployment ; decentralisation: shops, offices, leisure moving out of the CBD to out-of-town sites (e.g. retail parks, business parks) because of cheaper land and easier access ; CBD decline / empty shops, rise of online shopping ; suburbs / commuter towns expanded ;		4	AO1/AO2 3.2.1.3
03.7	car use fell from 48% to 27% ; cycling increased from 3% to 11% ; bus / rail use increased (14% → 25%, 12% → 22%) ; walking fell from 23% to 15% ;	Any two with data.	2	AO3 3.2.1.3
03.8	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO2 3.2.1.3
	Indicative content (credit any other valid geography, and place-specific detail): • named project and place: e.g. Queen Elizabeth Olympic Park, Stratford ; • reasons: derelict industrial / brownfield land, deprived area with high unemployment ; • social: new homes (affordable), schools, sports facilities (Aquatics Centre), the Athletes' Village converted to homes ; • economic: jobs in construction and later in retail (Westfield Stratford City), transport links (Jubilee line, DLR, Javelin) ; • environmental: cleanup of polluted land ; green space and wetlands ; improved river ; • evaluation: costs, rising rents / gentrification pushing out locals ;			
04.1	A: solar panels (on roof) ; C: trees / urban green space / park ;	B = (green / pitched) roof, D = recycling bin. Accept any reasonable naming of A and C.	2	AO2 3.2.1.4
04.2	meeting the needs of people today without harming the environment or the ability of future generations to meet their needs		1	AO1 3.2.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	water: rainwater harvesting / water meters / low-flush toilets / grey water recycling ; reduces demand for water and cost of treating it ; energy: solar panels / insulation / smart meters / low energy lighting ; reduces fossil fuel use and CO_2 emissions ; use of renewable energy (district heating) ;		4	AO1/AO2 3.2.1.4
04.4	trees absorb CO_2 / filter air pollution, giving cleaner air ; places for exercise and relaxation improve mental / physical health ; support wildlife / provide cooler temperatures (reduces urban heat island) ;		3	AO2 3.2.1.4
04.5	congestion charge / low emission zones: discourages car use, reducing traffic ; park and ride: cars left outside the centre, so fewer vehicles in the centre ; integrated public transport (buses, trams, cycle hire; e.g. Oyster) ; bus lanes and cycle lanes make alternatives faster and safer ; pedestrianised streets ;	Two strategies, each with 1 mark for the strategy and 1 for development.	4	AO2 3.2.1.4
04.6	**Level 3 (7–9 marks)**: Detailed, accurate knowledge and understanding of the topic with a well-developed, balanced and logical argument. A clear, supported **evaluation/judgement** that is based on evidence (and specific place detail or the resources where asked). Any conclusion is fully justified. **Level 2 (4–6 marks)**: Some accurate knowledge and understanding. Some development of ideas with evidence; the judgement/evaluation is partial, unbalanced or not fully supported. **Level 1 (1–3 marks)**: Limited knowledge and understanding. Simple statements with little development; a judgement, if any, is unsupported. **0 marks**: No relevant content.		9 + 3 SPaG	AO1/AO2/AO3 3.2.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
	Indicative content (credit any other valid geography, and place-specific detail): • Agree: fewer cars cut air pollution, CO_2 emissions, noise and congestion ; improves public health ; congestion charges raise money for public transport ; • Agree: promotes cycling and walking: healthier, quieter ; space freed for green areas ; • Disagree: other strategies are also important: energy conservation, recycling, renewable energy, green buildings, water conservation ; • Disagree/balance: car restrictions can harm businesses and be unfair to those in rural areas / disabled people without public transport ; need good public transport first ; • Evaluation: a combination of strategies (e.g. Curitiba, Freiburg) is likely to be most effective ; conclusion backed by named example ;			
	SPaG (3 marks) – award separately: **High performance (3)**: Learners spell and punctuate with consistent accuracy and consistently use rules of grammar; they use a wide range of specialist terms adeptly and with precision. **Intermediate performance (2)**: Learners spell and punctuate with considerable accuracy and use rules of grammar with general control; they use a good range of specialist terms with facility. **Threshold performance (1)**: Learners spell and punctuate with reasonable accuracy and use rules of grammar with some control; they use a limited range of specialist terms appropriately. **0 marks**: No marks awarded: no response or the spelling, punctuation and grammar is too weak, or the terminology is not relevant.			
05.1	City B	City B: 560 000 ÷ 42 000 = 13.3. City A = 6.9, City C = 6.4, City D = 9.7.	1	AO3 3.2.1.3
05.2	330 000 ÷ 34 000 ; 9.7 (allow 9.7 / 9.71) ;	Answer 9.7. Allow 10 for 1 mark if method shown.	2	AO4 3.2.1.3
05.3	greater demand / more people want to live there (jobs, culture) ; limited land supply / green belt prevents building ; higher wages in finance and services ; capital city attracts international buyers ;		2	AO2 3.2.1.3
05.4	brownfield sites: redevelop old factories / derelict land, so reuse of land ; build higher density / tower blocks (taller buildings), more homes on the same land ; convert disused offices and warehouses to flats ; infill developments / build in gardens ;	Two ways, each with 1 mark for the way and 1 for the reason.	4	AO2 3.2.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	loss of old industries / unemployment in inner cities ; older housing: poorer quality, overcrowded, more renting ; migrants arrive initially in cheaper inner city areas ; wealthier families move to suburbs, so remaining population has lower incomes ; cycle: poor services lead to poor health and education, so low incomes persist ;		4	AO2 3.2.1.3
05.6	growth of online shopping caused closures of high-street shops ; more cafés, restaurants, flats, and leisure uses instead of shops ; pedestrianisation / improved public spaces ; redevelopment: shopping centres and offices, e.g. Birmingham Bullring ;		2	AO1 3.2.1.3

Where marks are lost in Topic 4

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Urbanisation graphs:** quote figures from both lines and say which is higher / growing faster. Comparison words: 'whereas', 'compared with'.
- ☐ **Push–pull:** always give both: 'few jobs in rural areas (push) and better paid jobs in the city (pull)'.
- ☐ **Rio:** be specific – name Favela Bairro, Rocinha, Guanabara Bay, Copacabana, etc. A general answer about 'a city' limits you to the lower levels.
- ☐ **Social, economic, environmental:** split answers into these three headings and make sure each is covered, with a different example each time.
- ☐ **Case-study scheme:** 'aims → what was done → who benefits → what limits it'. A judgement ('mostly successful because…') earns the top level.
- ☐ **Deprivation data:** use 'highest', 'lowest', and numbers; calculate range = highest − lowest.
- ☐ **UK city:** know your example – London (migration, CBD, Stratford regeneration) with dates, numbers and named places.
- ☐ **9 marks:** plan a balanced answer with both views, named examples and a final judgement.